Proof-Obligation Cleanup for Systems Models

Abstract

Formal systems work becomes harder when the informal claim carries too much state at once. This manuscript studies a cleanup pass for systems models: separate the state, name the transitions, state the invariant, and keep each obligation close to the behavior it constrains.

Methodology

The workflow starts from a systems claim and extracts the objects it depends on: states, transitions, assumptions, invariants, and observable failures. Each object receives a narrow obligation. The paper then records which obligations can be discharged directly and which require changes to the model boundary.

Results

The resulting structure gives a readable proof plan for systems models. It keeps informal explanation, executable checks, and proof obligations aligned without turning the page into a proof dump.